S.NO.	Program name
1.	Write a Python program to perform arithmetic operations on integers.
2.	Write a Python program to check and print the types of at least 5 different inbuilt objects.
3.	Write a Python program to check if a number is EVEN or ODD.
4.	Write a Python program to check if a number is Positive, Negative or Zero.
5.	Write a Python program to check if a number is PRIME or NOT.
6.	Write a Python program to check whether a string entered by the user is a valid decimal number or not.
7.	Write a Python program to check if a year entered by the user is a Leap Year or NOT.
8.	Write a Python program to check whether a string entered by the user is a palindrome or not.
9.	Write a Python program to get a Decimal number from user and convert it into Binary, Octal and Hexadecimal.
10.	Write a Python program to find sum of natural numbers, up to N.
11.	Write a Python program to get marks in five subjects from user and calculate average marks, percentage and grade of a student.
12.	Write a Python program to get a number and find the sum and product of its digits.
13.	Write a Python program to get two integers and find their GCD and LCM.
14.	Write a Python program to find factorial of a number using while loop.
15.	Write a Python program to print Fibonacci series up to N terms.
16.	Write a Python program to print multiplication table.
17.	Write a Python program to access each element of a string in forward and backward orders using the 'while' loop.
18.	Write a Python program to access each element of a string in forward and backward orders using the 'for' loop.
19.	Write a Python program to find whether a substring exists in main string or not.
20.	Write a Python program to find the first occurrence of a substring in the main string.
21.	Write a Python program to convert from one number type to another.
22.	Write a Python program to demonstrate the use of all "casing" methods and display a string in different cases.
23.	write a Python program using a **while loop** to validate **User ID and Password** entered by the user.
24.	write a Python program using an **infinite `while` loop** to create a **menu-driven calculator**.
25.	write a Python program using a **for loop** to **count and display the number of vowels** in a given string.
26.	write a Python program using a loop that demonstrates the use of **`break`** and **`continue`** statements.
27.	write a Python program that uses a **ternary operator** to check a condition and return a value based on it.
28.	write a Python program that demonstrates the use of **`pass` statement** and **`print()` function parameters `end` and `sep`**.
29.	Write a Python Program to Create a dictionary of student names and marks. Traverse to display all students and find topper.
30	Write a Python Program to Search whether a roll number entered by the user exists in the list.
31.	Write a Python Program to Compare two names ignoring case sensitivity.

32.	Write a Python Program to Compare two strings and display which comes first alphabetically.
33.	Write a Python program to demonstrate the use of all “casing” methods and display a string in different cases.
34.	Write a Python program to demonstrate the use of all string testing {isXXX()} methods.
35.	Write a Python function to take a list of integers as input and return the average.
36.	Write a Python function to take two distinct integers as input and print all prime numbers between them.
37.	Write a Python function to take two integers as input and return both their sum and product.
38.	Write a Python program to demonstrate the positional arguments of a function.
39.	Write a Python program to demonstrate the keyword arguments of a function.
40.	Write a Python program to demonstrate the default arguments of a function.
41.	Write a Python function to demonstrate variable length arguments.
42.	Write a Python function to demonstrate keyword variable length arguments.
43.	Write a Python program to demonstrate global and local variables.

1. Write a Python program to perform arithmetic operations on integers.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
menu={
    "chai ":10,
    "coffee":20,
    "Cookeis":15
}
print(" Welcome to Red Brevarage Shop ")
print("Menu:")
for item,price in menu.items():
    print(f"{item}: ${price}")
total =0
while True:
    order = input("\nEnter item name (or 'exit' to finish): ").lower()

    if order == "exit":
        break

    if order in menu:
        qty = int(input("Enter quantity: "))
        total += menu[order] * qty
        print(f"{order} added to your order")
    else:
        print("Item not available")
print("\n Total Bill: ₹", total)
print("Thank you! Visit again ")
```

⇨ **Output**

```
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 Welcome to Red Brevarage Shop
Menu:
chai : $10
coffee: $20
Cookeis: $15

Enter item name (or 'exit' to finish): chai
Item not available

Enter item name (or 'exit' to finish): chai
Item not available

Enter item name (or 'exit' to finish): coffee
Enter quantity: 2
coffee added to your order

Enter item name (or 'exit' to finish): exit

 Total Bill: ₹ 40
Thank you! Visit again
```

2. Write a Python program to check and print the types of at least 5 different inbuilt objects.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

Name= " Swati yadav"
age=24
waight= 52.4
number ={"MCA":56}
verified=True
subject=["Python","DSA", "C++"]
print(type(Name))
print(type(age))
print(type(waight))
print(type(number))
print(type(verified))
print(type(subject))
```

⇨ **Output**

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program02.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
<class 'str'>
<class 'int'>
<class 'float'>
<class 'dict'>
<class 'bool'>
<class 'list'>
>>>
```

3. Write a Python program to check if a number is EVEN or ODD.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num=int(input(" Enter the number "))
if num%2==0:
    print(" this is even number  ")

else:
    print(" this is odd number ")
```

⇨ **Output**

```
>>>
================= RESTART: F:\python-ptc\PythonMCA\program03.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 Enter the number 18
 this is even number
>>>
```

4. Write a Python program to check if a number is Positive, Negative or Zero.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num=int(input("Enter your bank balence "))
if num>0:
   print(" Postive acoount ")
elif num==0:
   print("  yopur account is zero ")
else:
   print("your Account balence is  Negative ")
```

⇨ **Output**

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program04.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter your bank balence 150
 Postive acoount
```

5. Write a Python program to check if a number is PRIME or NOT.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num=int(input(" Enter your number: "))
if num<=1:
   print(" not prime ")
else:
   for i in range(2, num):
      if num % i == 0:
         print("Not Prime")
         break
   else
      print("Prime")
```

⇨ Output

```
>>>
================= RESTART: F:\python-ptc\PythonMCA\program05.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 Enter your number: 15
Not Prime
>>>
```

6. Write a Python program to check whether a string entered by the user is a valid decimal number or not.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
tapmaan= input(" temprature is : ")
if tapmaan.replace('.', '',1).isdigit():
   print("valid Decimal number:  ")
else:
   print(" Not valid ")
```

⇨ **Output**

```
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 temprature is : 35.6
valid Decimal number:
```

6. Write a Python program to check if a year entered by the user is a Leap Year or NOT.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
year = int(input(" Happy new Year : "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
   print("leap year")
else:
   print(" not leap year ")
```

⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program07.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 Happy new Year : 2026
 not leap year
```

8. Write a Python program to check whether a string entered by the user is a palindrome or not.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

alphabet = input(( "Enter your words" ))
if alphabet==alphabet[:: -1]:
   print(" Palindrome ")
else:
   print(" not palindrom ")
```

⇨ Output

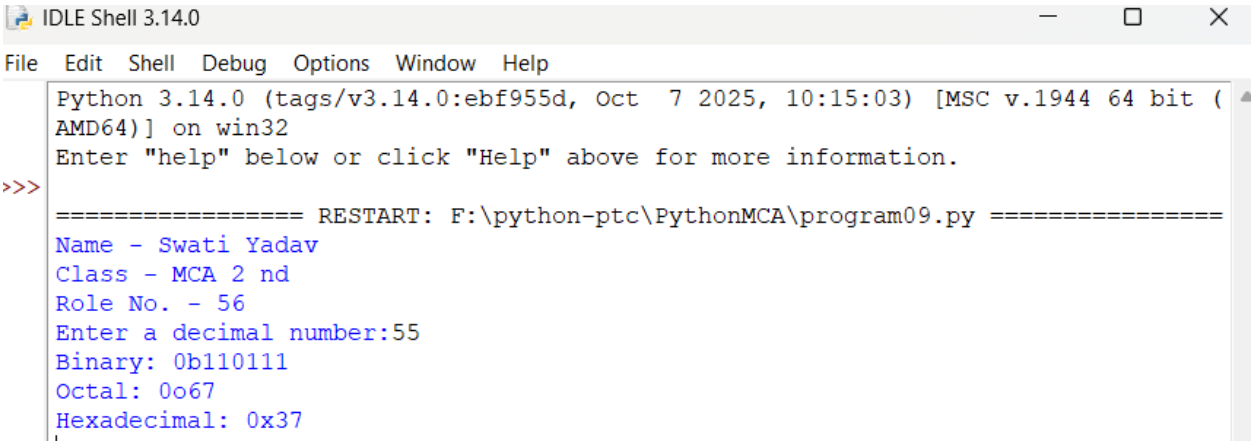
```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
================= RESTART: F:\python-ptc\PythonMCA\program08.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter your words Belive
 not palindrom
>>
```

9. Write a Python program to get a Decimal number from user and convert it into Binary, Octal and Hexadecimal.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

num = int(input("Enter a decimal number:"))

print("Binary:", bin(num))
```

```
print("Octal:", oct(num))
print("Hexadecimal:", hex(num))
```

⇨ **Output**



10. Write a Python program to find sum of natural numbers, up to N.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

n = int(input("Enter a number: "))

total = 0
for i in range(1, n + 1):
```

```
    total += i

print("Sum:", total)
```

⇨ **Output**

```
AMD64)] on win32
Enter "help" below or click "Help" above for more infor

================= RESTART: F:\python-ptc\PythonMCA\prog
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a number: 56
Sum: 1596
```

11. Write a Python program to get marks in five subjects from user and calculate average marks, percentage and grade of a student.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

name=input("Enter your name : ")
sub1 = float(input("Enter marks for Subject 1: "))
```

```
sub2 = float(input("Enter marks for Subject 2: "))
sub3 = float(input("Enter marks for Subject 3: "))
sub4 = float(input("Enter marks for Subject 4: "))
sub5 = float(input("Enter marks for Subject 5: "))

total = sub1 + sub2 + sub3 + sub4 + sub5

average = total / 5

percentage = (total / 500) * 100

if percentage>60:
   print("Grade A")
elif percentage>40:
   print("Grade B")
else:
   print(" fail ")
print("Total Marks:", total)
print("Average Marks:", average)
print("Percentage:", percentage, "%")
print("Student Name : " ,name)
```

⇨ **Output**

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: F:\python-ptc\PythonMCA\program11.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter your name : swati yadav
Enter marks for Subject 1: 65
Enter marks for Subject 2: 58
Enter marks for Subject 3: 74
Enter marks for Subject 4: 89
Enter marks for Subject 5: 98
Grade A
Total Marks: 384.0
Average Marks: 76.8
Percentage: 76.8 %
Student Name :  swati yadav
>>>
```

12. Write a Python program to get a number and find the sum and product of its digits.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num = int(input(" Enter the number : "))
```

```
sum_digits=0
product=1

while num>0:
    digit=num%10
    sum_digit+=digit
    product *=digit
    num = num//10

print("Sum:", sum_digits)
print("Product:", product)
```

⇨ **Output**

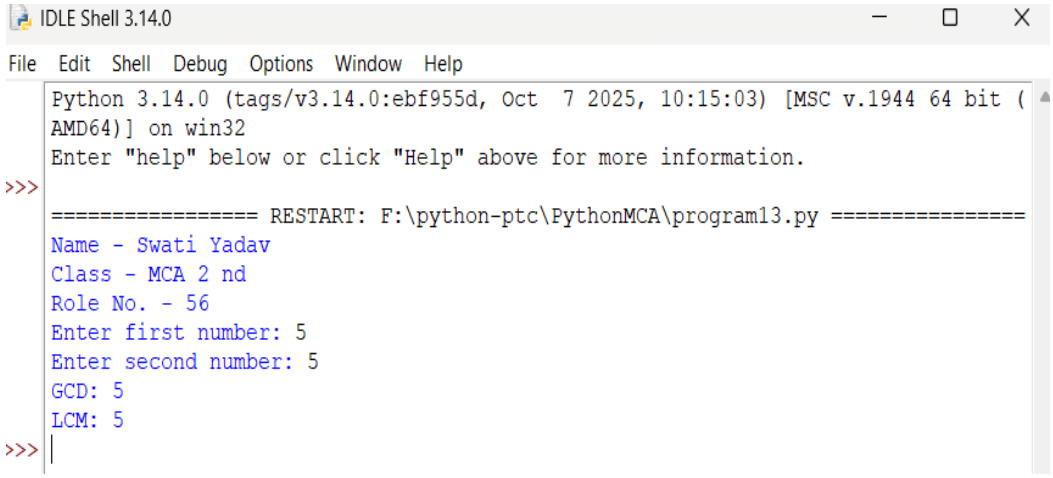
```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program12.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
 Enter the number : 45
Sum: 9
Product: 20
>>>
```

13. Write a Python program to get two integers and find their GCD and LCM.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
```

```
print("Role No. - 56")

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

x, y = a, b
while y != 0:
    x, y = y, x % y
gcd = x

lcm = (a * b) // gcd

print("GCD:", gcd)
print("LCM:", lcm)
```

⇨ **Output**

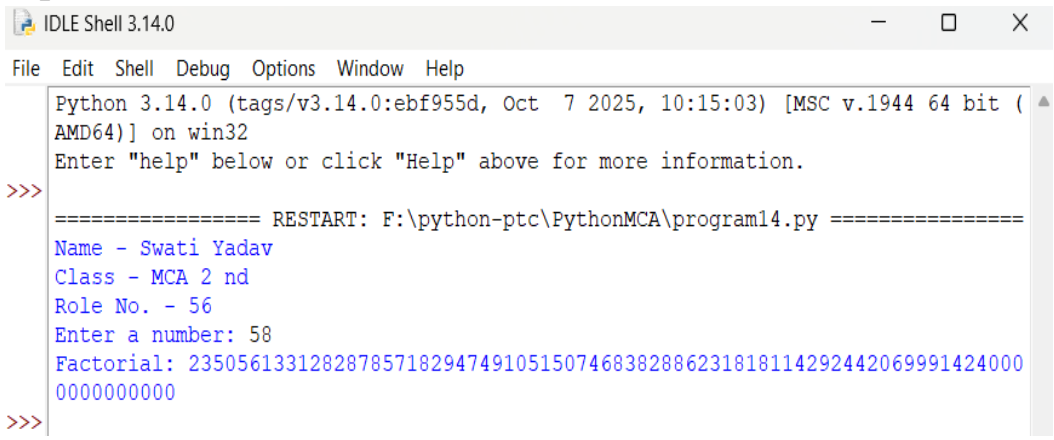


14. Write a Python program to find factorial of a number using while loop.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
```

```
print("Role No. - 56")

num = int(input("Enter a number: "))

fact = 1
i = 1

while i <= num:
    fact *= i
    i += 1

print("Factorial:", fact)
```

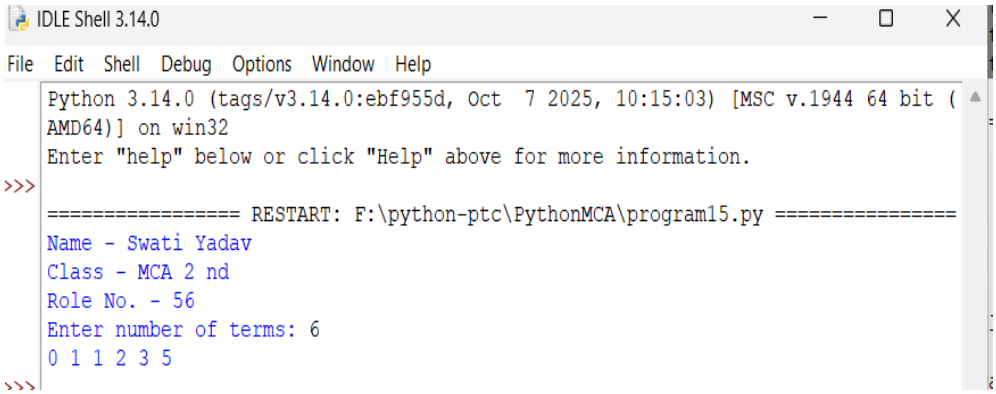
⇨ **Output**



15. Write a Python program to print Fibonacci series up to N terms.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

n = int(input("Enter number of terms: "))

a, b = 0, 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

⇨ **Output**



16. Write a Python program to print multiplication table.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

num = int(input("Enter a number: "))

for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program16.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a number: 45
45 x 1 = 45
45 x 2 = 90
45 x 3 = 135
45 x 4 = 180
45 x 5 = 225
45 x 6 = 270
45 x 7 = 315
45 x 8 = 360
45 x 9 = 405
45 x 10 = 450
>>>
```

17. Write a Python program to access each element of a string in forward and backward orders using the `while` loop.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
name = input("Enter a string: ")

# Forward
i = 0
while i < len(name):
    print(name[i], end=" ")
    i += 1

print()

# Backward
i = len(name) - 1
while i >= 0:
    print(name[i], end=" ")
    i -= 1
```

⇨ Output

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program17.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a string: Won
W o n
n o W
>>>
```

18. Write a Python program to access each element of a string in forward and backward orders using the `for` loop.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
name = input("Enter a string: ")

for ch in name:

    print(ch, end=" ")

print()

# Backward
for i in range(len(name) - 1, -1, -1):
    print(name[i], end=" ")
```

⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program18.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a string: bye
b y e
e y b
>>>
```

19. Write a Python program to find whether a substring exists in main string or not.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

main_str = input("Enter main string: ")
sub_str = input("Enter substring: ")

if sub_str in main_str:
    print("Substring exists")
else:
    print("Substring does not exist")
```

⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: F:\python-ptc\PythonMCA\program18.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a string: bye
b y e
e y b
>>>
```

20. Write a Python program to find the first occurrence of a substring in the main string.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num = input("Enter a number: ")
a = int(num)

b = float(a)

c = str(b)

print(a, type(a))
print(b, type(b))
```

```
print(c, type(c))
```

⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: F:\python-ptc\PythonMCA\program19.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter main string: main event
Enter substring: WWE
Substring does not exist
>>>
```

21. Write a Python program to convert from one number type to another.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num = input("Enter a number: ")

# Convert string to int
a = int(num)

# Convert int to float
b = float(a)
```

```
# Convert float to string
c = str(b)

print(a, type(a))
print(b, type(b))
print(c, type(c))
```

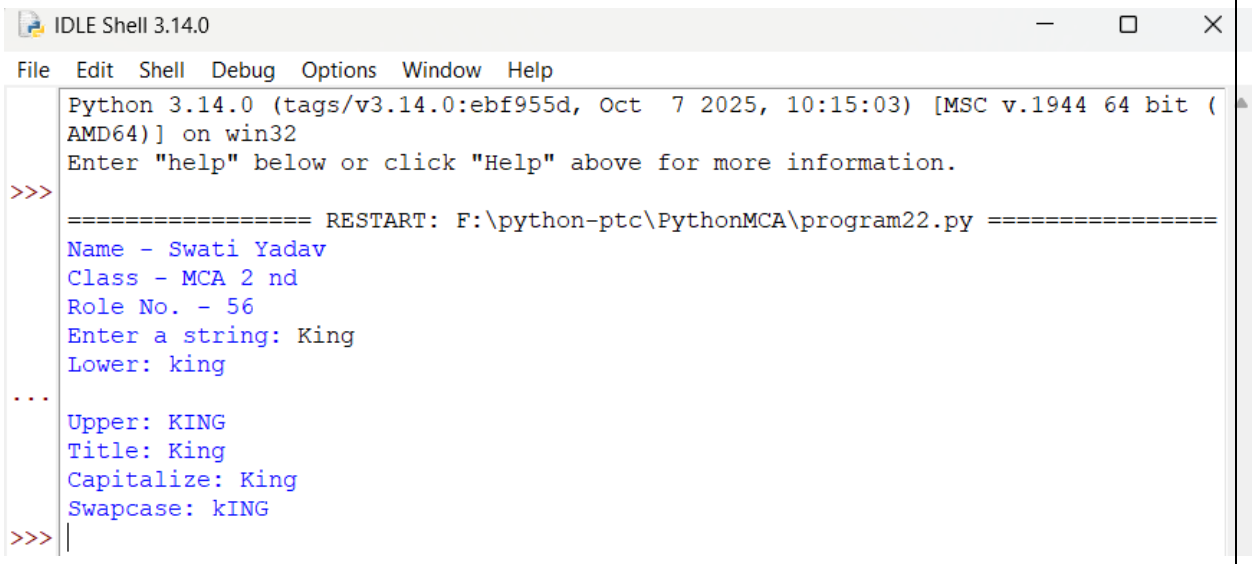
⇨ Output

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program21.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter a number: 56
56 <class 'int'>
56.0 <class 'float'>
56.0 <class 'str'>
>>>
```

22. Write a Python program to demonstrate the use of all "casing" methods and display a string in different cases.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
s = input("Enter a string: ")

print("Lower:", s.lower())
print("Upper:", s.upper())
print("Title:", s.title())
```

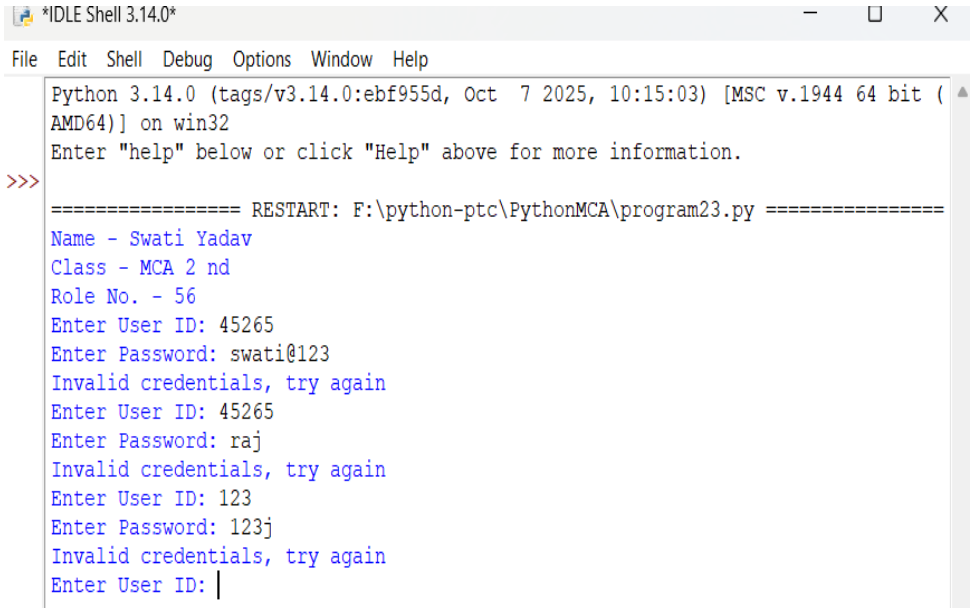
```
print("Capitalize:", s.capitalize())
print("Swapcase:", s.swapcase())
```



23. Write a Python program using a while loop to validate User ID and Password entered by the user.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
correct_id = "admin"
correct_pass = "admin123"

while True:
```

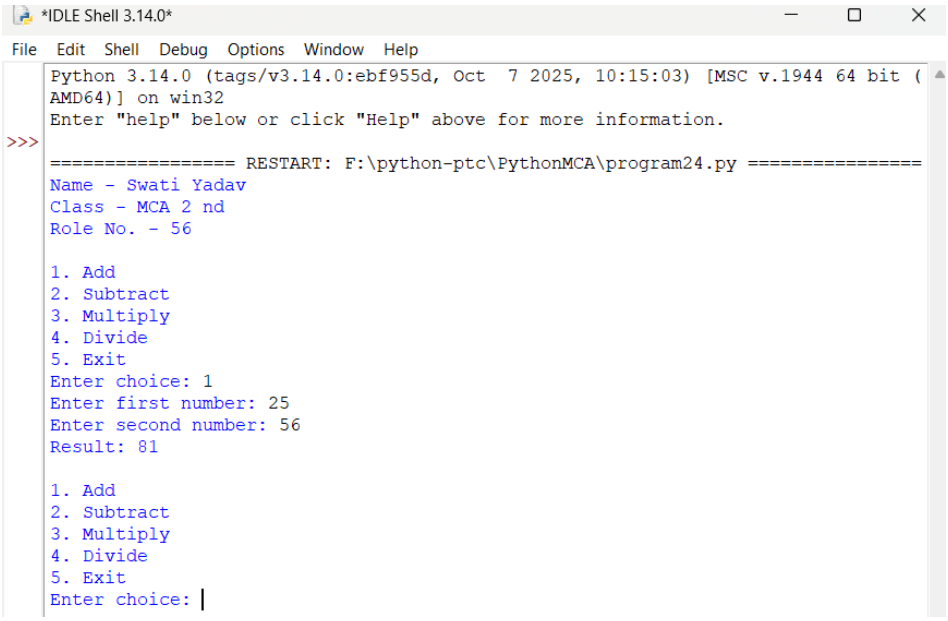
```
    user_id = input("Enter User ID: ")
    password = input("Enter Password: ")

    if user_id == correct_id and password == correct_pass:
        print("Login Successful")
        break
    else:
        print("Invalid credentials, try again")
```



24. Write a Python program using an infinite while loop to create a menu-driven calculator.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

while True:
```

```
print("\n1. Add\n2. Subtract\n3. Multiply\n4. Divide\n5. Exit")
choice = int(input("Enter choice: "))

if choice == 5:
    print("Exiting...")
    break

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if choice == 1:
    print("Result:", a + b)
elif choice == 2:
    print("Result:", a - b)
elif choice == 3:
    print("Result:", a * b)
elif choice == 4:
    if b != 0:
        print("Result:", a / b)
    else:
        print("Cannot divide by zero")
else:
    print("Invalid choice")
```

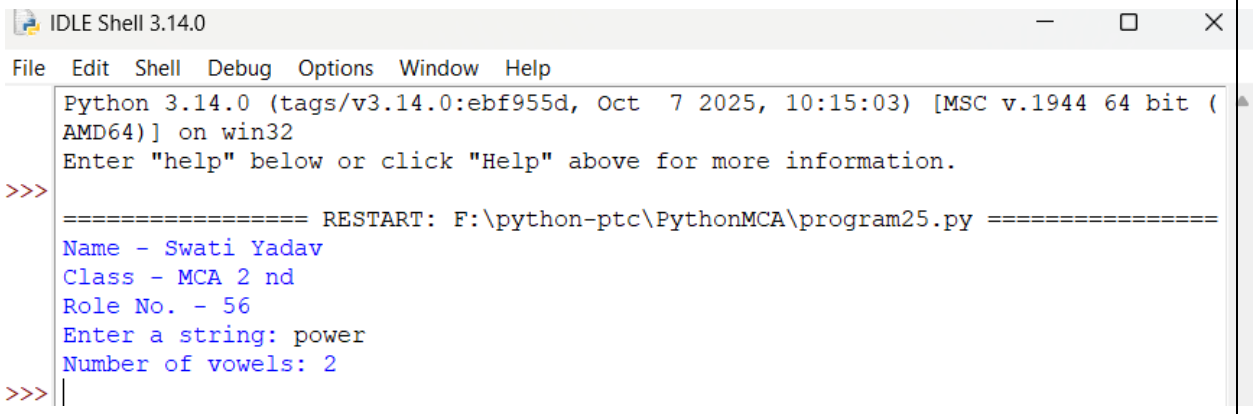


25. Write a Python program using a for loop to count and display the number of vowels in a given string.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

s = input("Enter a string: ")
count = 0

for ch in s:
    if ch.lower() in "aeiou":
```

```
        count += 1

print("Number of vowels:", count)
```

⇨ Output



26. Write a Python program using a loop that demonstrates the use of `break` and `continue` statements.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

for i in range(1, 10):
    if i == 3:
        print("Skipped 3")
        continue
    if i == 6:
        print("Loop stopped at 6")
```

```
        break
    print(i)
```

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program26.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
1
2
Skipped 3
4
5
Loop stopped at 6
>>>
```

27. Write a Python program that uses a ternary operator to check a condition and return a value based on it.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
num = int(input("Enter number: "))
result = "Even" if num % 2 == 0 else "Odd"
print(result)
```

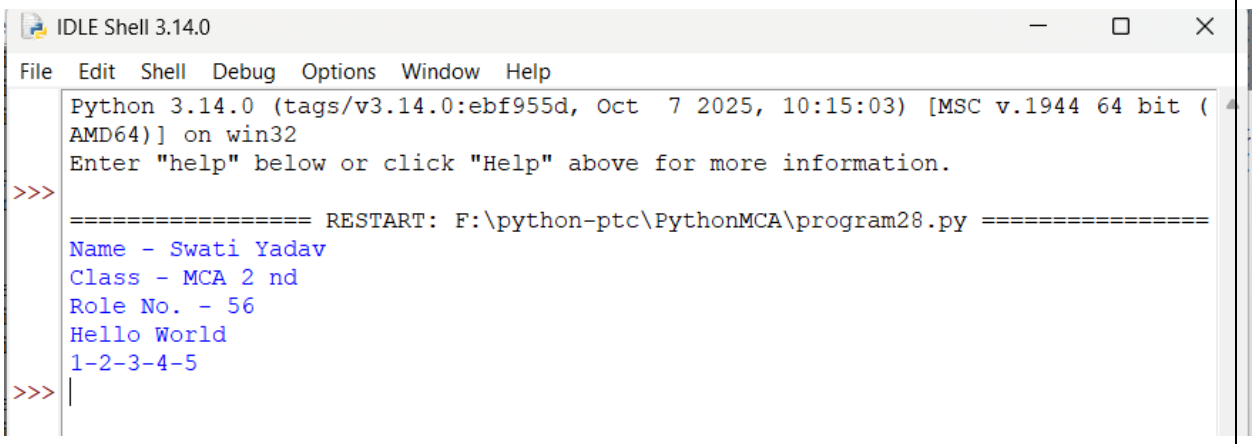
⇨ **Output**

```
IDLE Shell 3.14.0
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program27.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Enter number: 125
Odd
>>>
```

28. Write a Python program that demonstrates the use of `pass` statement and `print()` function parameters `end` and `sep`.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
```

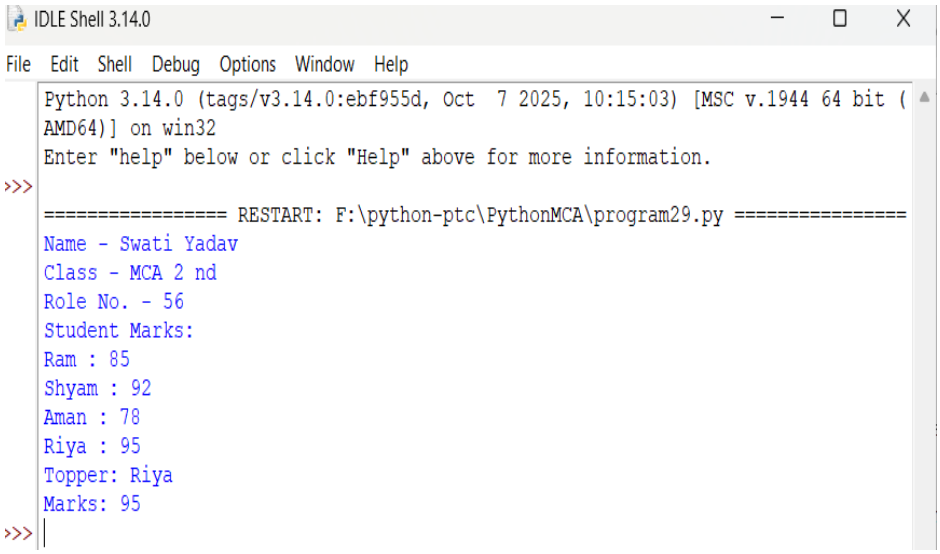
```
# pass example
for i in range(3):
    pass  # does nothing

# end parameter
print("Hello", end=" ")
print("World")

# sep parameter
print(1, 2, 3, 4, 5, sep="-")
```



29. Write a Python Program to create a dictionary of student names and marks. Traverse to display all students and find topper.

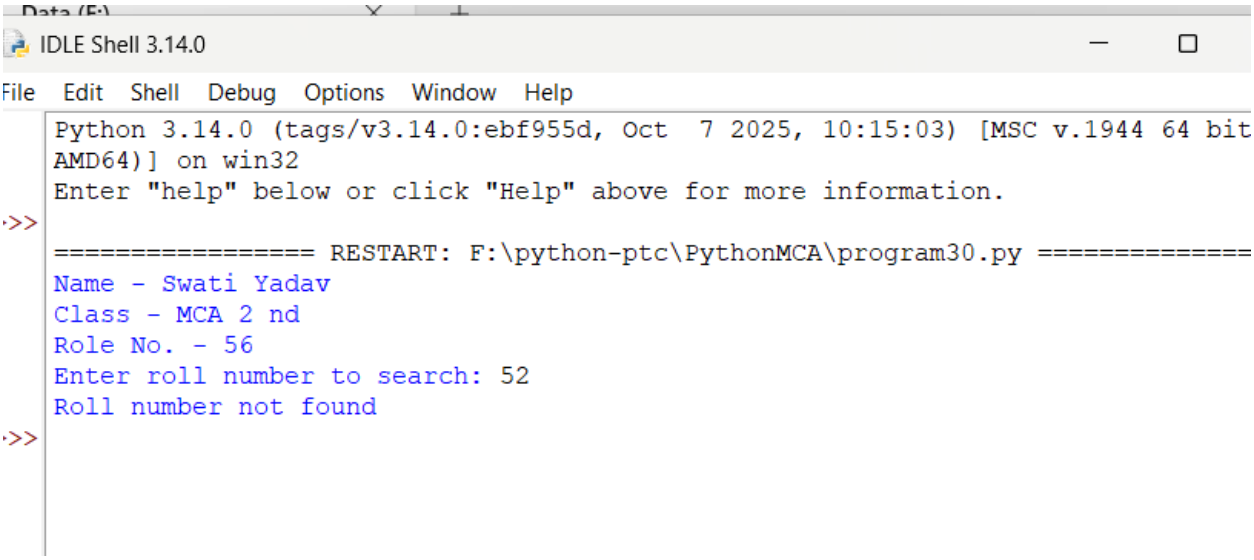
```
print("Name - Swati Yadav ")
```

```
print("Class - MCA 2 nd ")
print("Role No. - 56")
students = {
    "Ram": 85,
    "Shyam": 92,
    "Aman": 78,
    "Riya": 95
}

print("Student Marks:")
for name, marks in students.items():
    print(name, ":", marks)

topper = max(students, key=students.get)

print("Topper:", topper)
print("Marks:", students[topper])
```



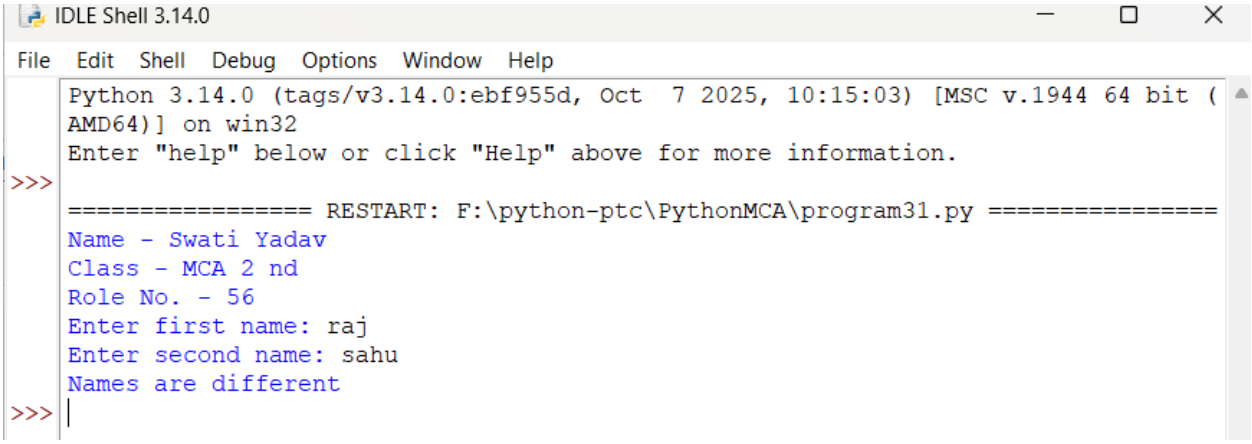
30. Write a Python Program to search whether a roll number entered by the user exists in the list.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
roll_numbers = [101, 102, 103, 104, 105]

num = int(input("Enter roll number to search: "))

if num in roll_numbers:
    print("Roll number exists")
else:
    print("Roll number not found")
```



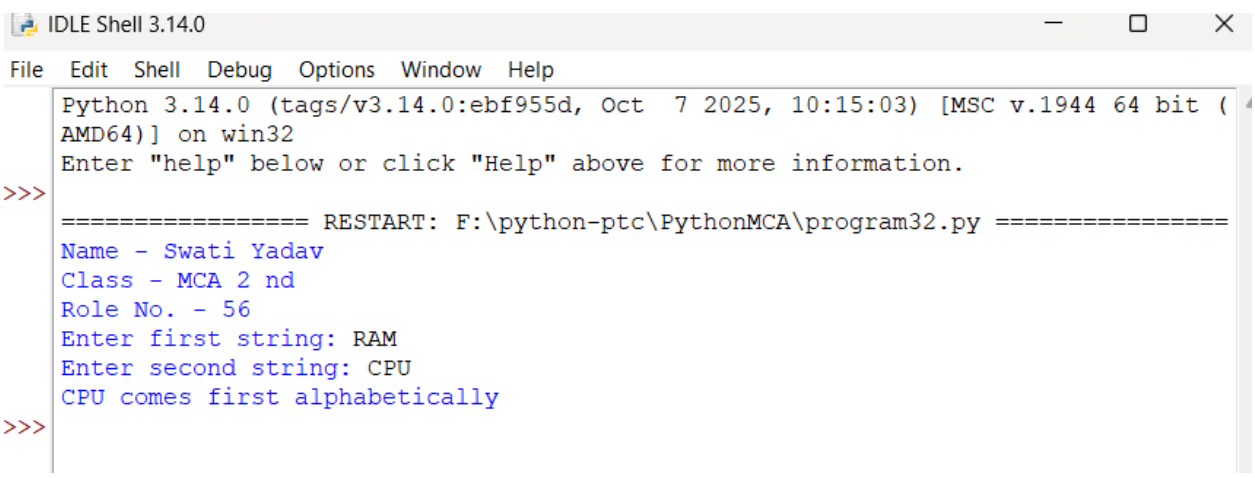
31. Write a Python Program to compare two names ignoring case sensitivity.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

name1 = input("Enter first name: ")
name2 = input("Enter second name: ")

if name1.lower() == name2.lower():
    print("Both names are equal")
else:
    print("Names are different")
```



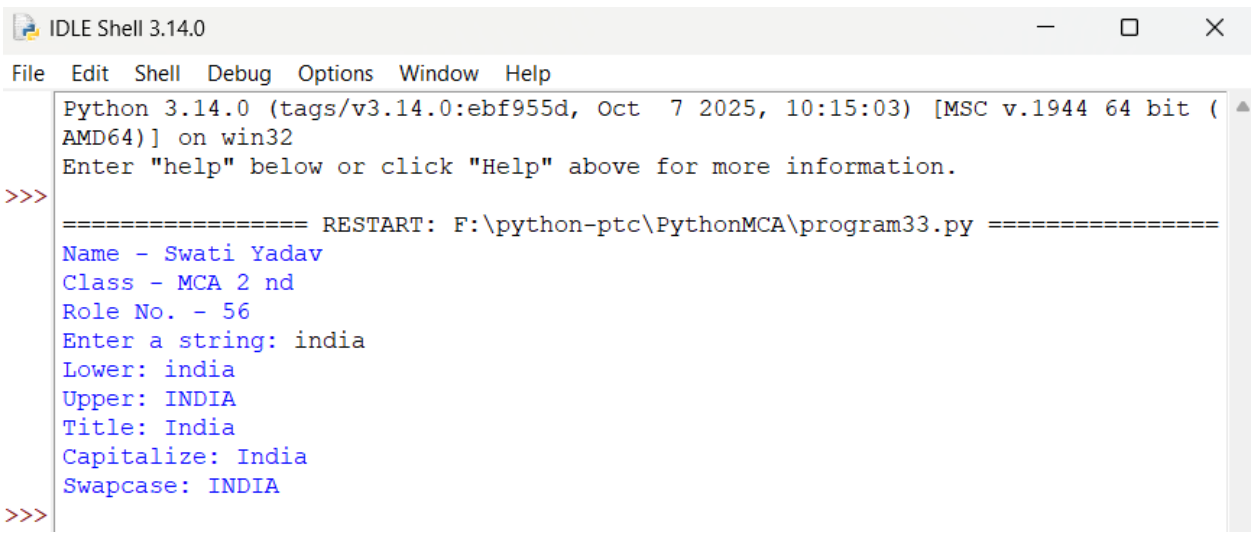
32. Write a Python Program to compare two strings and display which comes first alphabetically.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if str1 < str2:
    print(str1, "comes first alphabetically")
elif str1 > str2:
    print(str2, "comes first alphabetically")
else:
    print("Both strings are equal")
```



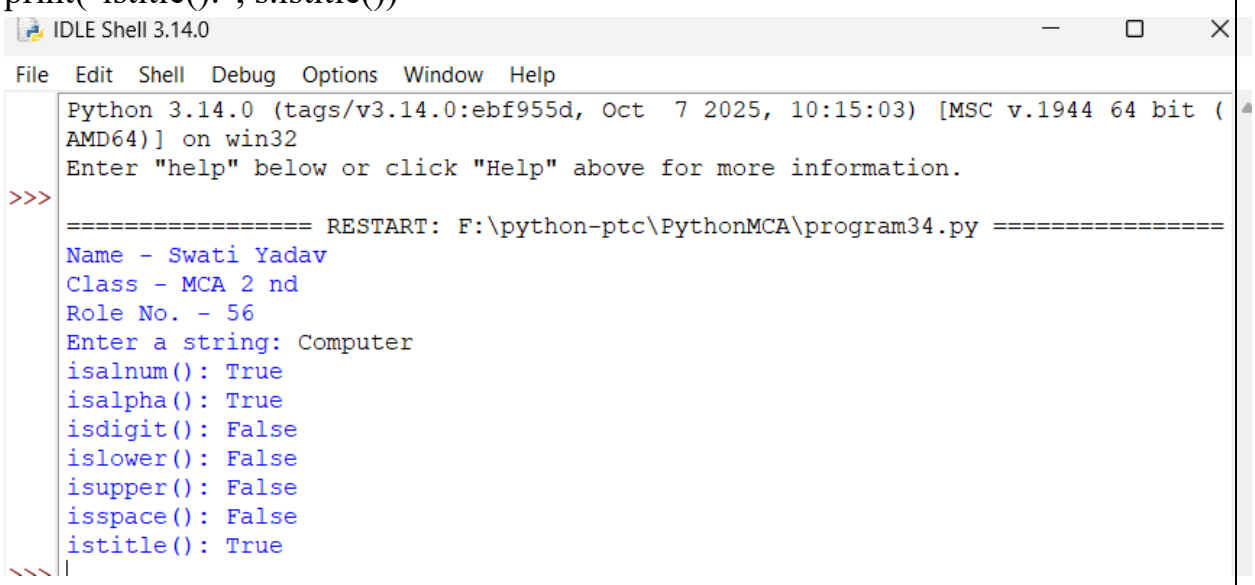
33. Write a Python program to demonstrate the use of all “casing” methods and display a string in different cases.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

s = input("Enter a string: ")

print("Lower:", s.lower())
print("Upper:", s.upper())
print("Title:", s.title())
print("Capitalize:", s.capitalize())
print("Swapcase:", s.swapcase())
```



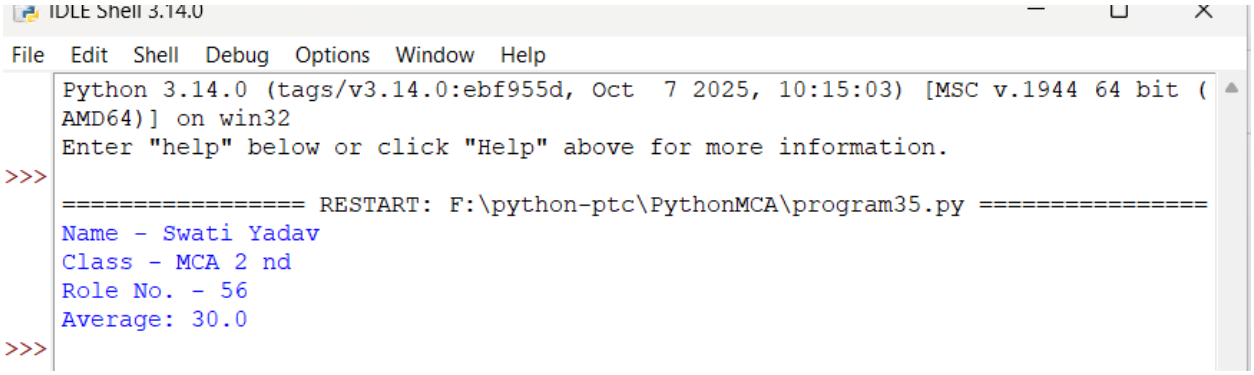
34. Write a Python program to demonstrate the use of all string testing {isXXX()} methods.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")


s = input("Enter a string: ")

print("isalnum():", s.isalnum())
print("isalpha():", s.isalpha())
print("isdigit():", s.isdigit())
print("islower():", s.islower())
print("isupper():", s.isupper())
print("isspace():", s.isspace())
print("istitle():", s.istitle())
```



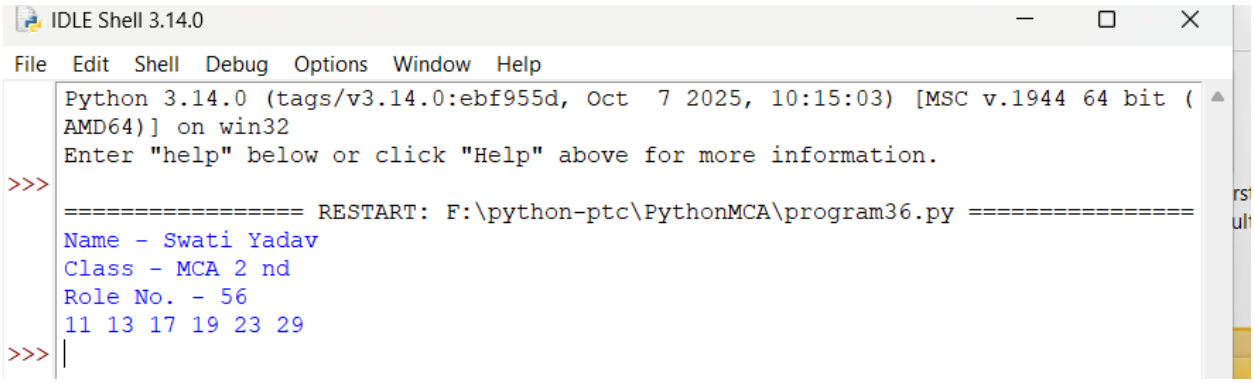
35. Write a Python function to take a list of integers as input and return the average.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
def find_average(numbers):
    total = sum(numbers)
    avg = total / len(numbers)
    return avg

# Example
nums = [10, 20, 30, 40, 50]
print("Average:", find_average(nums))
```



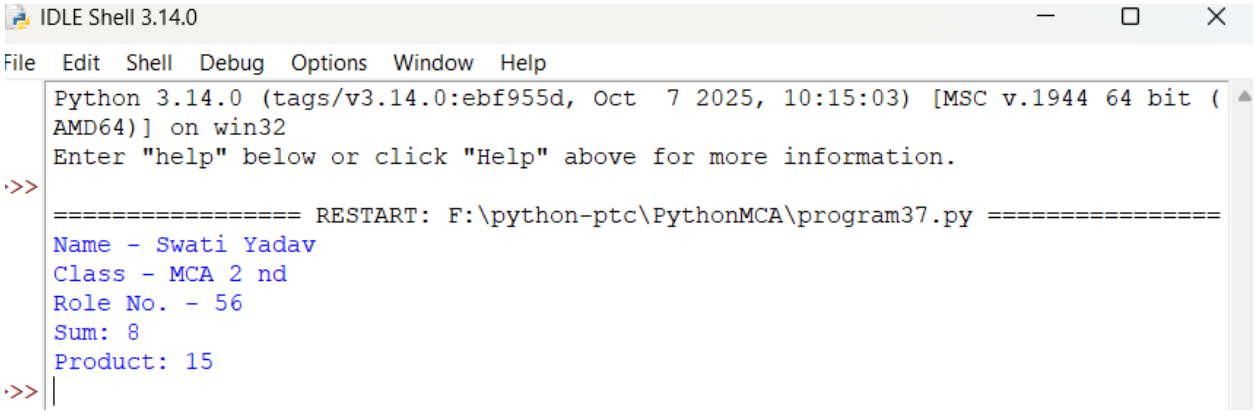
36. Write a Python function to take two distinct integers as input and print all prime numbers between them.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")


def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

def print_primes(a, b):
    for num in range(a, b + 1):
        if is_prime(num):
            print(num, end=" ")

print_primes(10, 30)
```



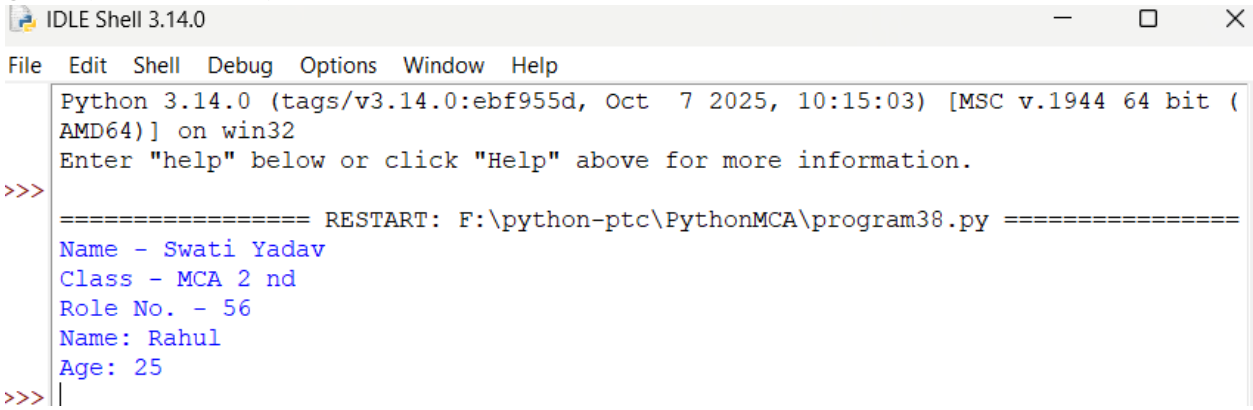
37. Write a Python function to take two integers as input and return both their sum and product.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

def sum_and_product(a, b):
    return a + b, a * b

# Example
s, p = sum_and_product(5, 3)
print("Sum:", s)
print("Product:", p)
```



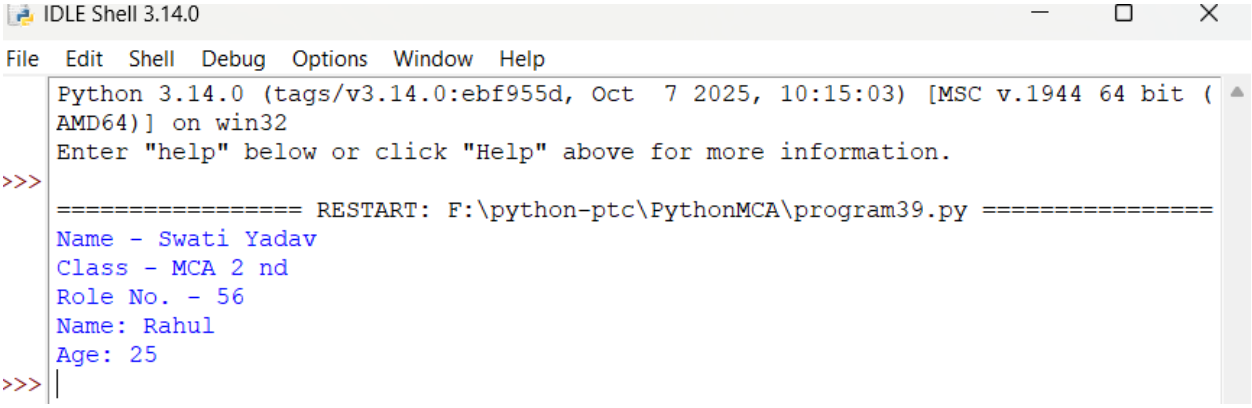
38. Write a Python program to demonstrate the positional arguments of a function.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
def greet(name, age):
    print("Name:", name)
    print("Age:", age)

greet("Rahul", 25)
```



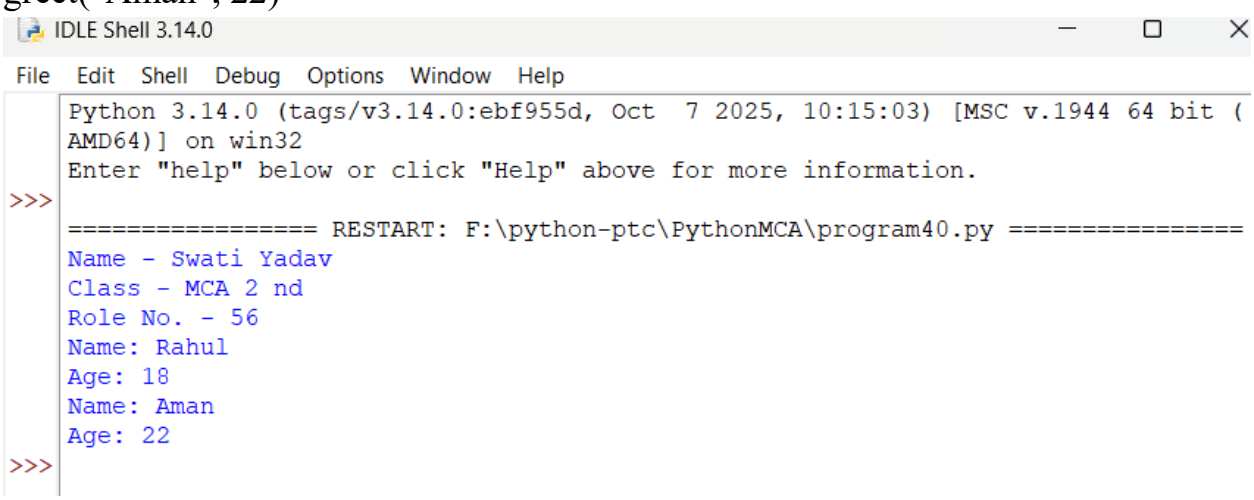
39. Write a Python program to demonstrate the keyword arguments of a function.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")
def greet(name, age):
    print("Name:", name)
    print("Age:", age)


greet(age=25, name="Rahul")
```



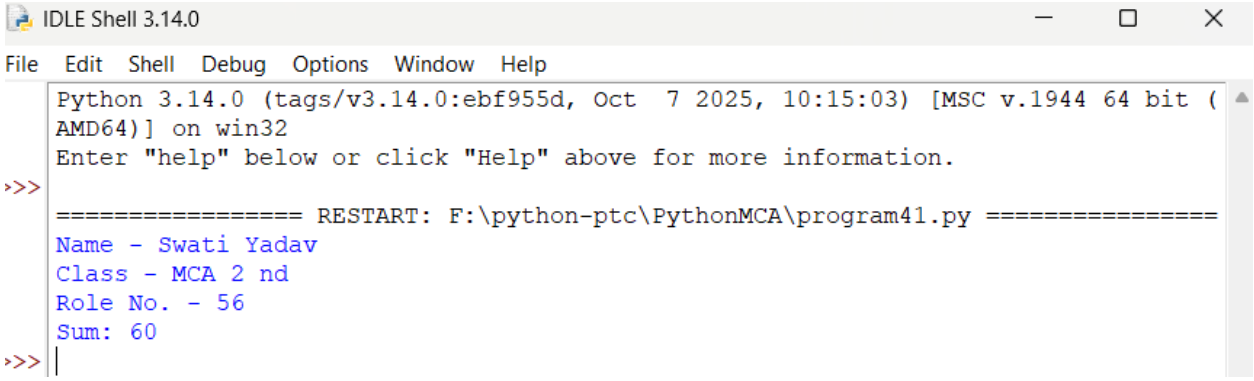
40. Write a Python program to demonstrate the default arguments of a function.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")


def greet(name, age=18):
    print("Name:", name)
    print("Age:", age)

# Using default age
greet("Rahul")

# Overriding default age
greet("Aman", 22)
```



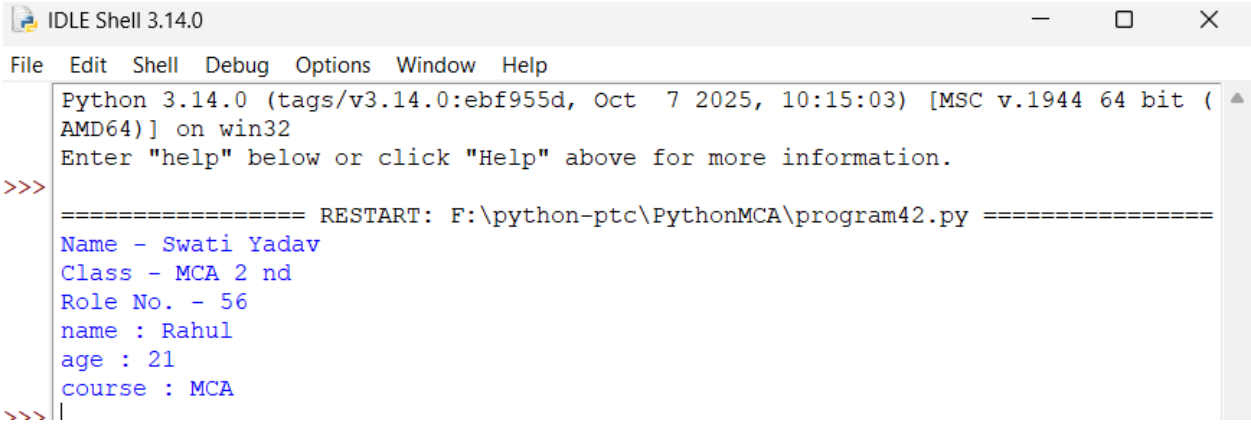
41. Write a Python function to demonstrate variable length arguments.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

def add_numbers(*args):
    total = 0
    for num in args:
        total += num
    print("Sum:", total)

# Passing multiple values
add_numbers(10, 20, 30)
```



42. Write a Python function to demonstrate keyword variable length arguments.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

def student_info(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

# Passing key-value pairs
student_info(name="Rahul", age=21, course="MCA")
```



43. Write a Python program to demonstrate global and local variables.

```
print("Name - Swati Yadav ")
print("Class - MCA 2 nd ")
print("Role No. - 56")

x = 10

def show():

    y = 5
    print("Inside function:")
    print("Global x =", x)
    print("Local y =", y)

show()

print("\nOutside function:")
print("Global x =", x)
```

⇨ Output

IDLE Shell 3.14.0

File Edit Shell Debug Options Window Help

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: F:\python-ptc\PythonMCA\program43.py ================
Name - Swati Yadav
Class - MCA 2 nd
Role No. - 56
Inside function:
Global x = 10
Local y = 5

Outside function:
Global x = 10
>>>
```